Math708 – Foundations of Computational Mathematics I

Fall 2012

Midterm Exam 1

You must show all of your work to get a credit for a correct answer.

1. (20 points) Write down the Lagrange interpolation polynomial $p_1(x) \in \mathbb{P}_1$ for the function $f : x \to (2x-a)^4$ using the points $x_0 = 0$, $x_1 = a$. Show that there are two possible values for ξ in the interpolation error $f(x) - p_1(x) = \frac{\omega_2(x) f''(\xi)}{2!}$ and give their values.

2. (20 points) Let $(x_0, f_0), (x_1, f_1), \cdots, (x_n, f_n) \in \mathbb{R}^2$ and let $(y_0, g_0), (y_1, g_1), \cdots, (y_n, g_n) \in \mathbb{R}^2$ be support points with corresponding divided differences $f[x_0, \cdots, x_n]$ and $g[y_0, \cdots, y_n]$. Prove that if $\{(x_k, f_k), k = 0, 1, \cdots, n\} = \{(y_k, g_k), k = 0, 1, \cdots, n\}$, then $f[x_0, \cdots, x_n] = g[y_0, \cdots, y_n]$.

3. (20 points) Given the following data

x_n	0	1	2	4	6
$f(x_n)$	1	9	23	93	259

   Construct a Newton's interpolation polynomial that assumes these values using divided differences. Then write the polynomial obtained above in nested form for easy evaluation (Horner's algorithm).

4. (20 points) Given the following data

x_n	-1	0	$\frac{1}{2}$	1
$f(x_n)$	2	1	0	1

   Construct a quadratic spline s_2 interpolating these data with additionally $s_2'(-1) = 0$.

5. (20 points) Let $N \in \mathbf{N}$ be even number. For $\{f_k\}_{k=0}^{N-1} \in \mathbb{R}^N$, suppose that $f_{N-k} = f_k$ for $k = 1, \cdots, N/2 - 1$. Show that:

$$d_j \in \mathbb{R} \quad \text{for} \quad j = 0, \cdots, N-1, \quad \text{and} \quad d_{N-j} = d_j \quad \text{for} \quad j = 1, \cdots, N-1.$$